

Zhengning Xu

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Research Interests

AI4S: computational biology, multimodal animal behavior, generative world models and causal video evaluation; model generalization, interpretability, and scientific agents.

Education

ShanghaiTech University

Sep 2024–Present

B.S. candidate in Computer Science and Technology | SIST | AI Honors Program (selected 2025)

- GPA **3.89/4.0**; weighted average **94.21/100**; ranked **4/135** in Computer Science and Technology and **5/260** in SIST, as of Sep 2026.
- Relevant coursework: Bioinformatics: Software Development and Applications (A+), Algorithms and Data Structures (A+), Discrete Mathematics (A+), Probability and Statistics for Information Science (A), Signals and Systems (A+), Electrophysiology and Brain-Machine Interfaces (A+), and Computer Architecture I (A).

Honors and Awards

- Outstanding Student and Outstanding Member of the Communist Youth League, ShanghaiTech University, 2025.
- Outstanding Trainee, Oriental Land Military Training, ShanghaiTech University, 2024.
- Outstanding Individual, Undergraduate Social Practice (2025) and Industry Practice (2026), ShanghaiTech University.
- Outstanding Student Group Leader, Shangdao College, ShanghaiTech University, 2024 and 2025.
- AI Honors Program (Honors Degree), School of Information Science and Technology, ShanghaiTech University.

Research Experience

DeepDeltaLab, School of Artificial Intelligence, Shanghai Jiao Tong University

Feb 2026–Present

Undergraduate Research Intern | Advisor: Prof. Shaobo Cui

- Co-designed a cancer drug-response model by comparing cell-line representations (gene expression and pathway activity) with molecular sequence and 2D/3D chemical representations; contributed to joint drug–cell-line prediction, feature encoding, and training interfaces.
- Built a controlled benchmark and interpretability workflow: reconciled drug and cell-line identifiers, fixed data splits, retrained public baselines, and examined gene attributions through virtual knockouts, pathway analysis, and independent literature checks across training folds.
- Led the end-to-end VQA evaluation of generated videos for CausalCascade and wrote the manuscript; designed event-graph questions, assessed event occurrence and temporal dependencies, and analyzed failures under counterfactual interventions.

MouseGPT Team, ShanghaiTech University

Sep 2025–Jan 2026

Undergraduate Researcher | Advisors: Prof. Jingyi Yu and Prof. Ji Hu | Paper coauthor

- Reproduced the mouse-video analysis pipeline: deployed the vision-language model, connected video frames, pose statistics, behavior descriptions, and semantic embeddings, and integrated DeepLabCut, SLEAP, and Keypoint-MoSeq for pose and behavior analysis comparisons.
- Built cross-dataset evaluations on CalMS21, MABE 2022, and multiview data, comparing description quality, behavior classification, and view consistency with expert blind review and conventional pose-based methods.
- Improved the downstream agent interface linking natural-language requests to behavior retrieval, statistics, and visualization; tested intent interpretation, tool calls, and result validity across datasets.

Publications and Manuscripts

- **MouseGPT: A Large-scale Vision-Language Model for Mouse Behavior Analysis.** Coauthor. Minor revision at *Nature Neuroscience*, 2026.
- **A Plausible Future Is Not a Causal Future: Can Video Models Generate How the World Unfolds?** Co-first author. Submitted to ICLR 2027.
- **A Wearable Fabric Wristband System for Smart and Comfortable Human-machine Interaction.** Coauthor. Accepted to IEEE ROBIO 2025.

Selected Projects

Context-Specific Synthetic Lethality Prediction

Feb–Jun 2025

CS177 Bioinformatics: Software Development and Applications, ShanghaiTech University | Grade: A+

- Designed a data pipeline joining SLKB, DepMap, and STRING labels, cell-line expression, and protein interaction graphs; combined scGPT, GenePT, ESM-2, and cell-specific PPI features in a GCN/MLP model for gene-pair prediction.
- Reproduced KG4SL and MVGCN-iSL and designed within-cell-line, cross-cell-line, and modality ablation experiments; examined what expression, protein sequence, and interaction networks contribute to predictions in unseen cellular contexts.

CausalCascade: Evaluating Causal Processes in Video World Models

2026

DeepDeltaLab, School of Artificial Intelligence, Shanghai Jiao Tong University | Submitted to ICLR 2027

- Built an event-graph-guided VQA evaluation workflow for generated videos across 50 physical scenes, 2,832 controlled variants, and paired counterfactual interventions; checked event occurrence, temporal dependencies, and causal end states.
- Analyzed nine image-to-video models on event understanding, motion localization, and responses to interventions; identified gaps between recognizing likely events and generating complete physical processes, and wrote the manuscript.

Fabric Wristband for Physiological and Motion Sensing

2025

ShanghaiTech University PRP | Paper accepted to IEEE ROBIO 2025

- The team developed a wearable system combining breathable fabric, liquid-metal printed circuits, PPG, and a nine-axis IMU for comfortable physiological sensing and gesture recognition.
- Developed the signal-processing pipeline: estimated heart rate and oxygen saturation from PPG, calibrated and cleaned IMU sequences, and combined dynamic time warping with k-means to distinguish forehand and backhand swings across speeds.

Campus Service and Practice

- Student delegate to the Sixth ShanghaiTech University Student Congress in Oct 2025; recognized twice as Outstanding Student Group Leader at Shangdao College.
- Recognized as Outstanding Individual in ShanghaiTech’s Undergraduate Social Practice (2025) and Industry Practice (2026).

Technical Skills

Programming	Python, C/C++; PyTorch, Git, Docker, Linux, $\LaTeX$.
Methods	Multimodal representations, graph neural networks, vision-language models, controlled benchmarking, model attribution, and literature evidence audits.
English	College English Test Band 6 (CET-6): 576, Dec 2025.